



Hibernate 3.0

*A discussion of the new features in Hibernate 3.0,
and how to take advantage of them.*

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Overview

- **Hibernate is an open-source Object-Relational Mapping (ORM) tool**
- **First release (0.8) November 2001**
 - 1.0 release July 2002**
 - 2.0 release June 2003**
 - 3.0 release Mar 2005**

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New Features

- **XML Mappings**
- **Bulk update/delete by query**
- **Virtualisation**
- **Single-object to multiple-table mapping**
- **Support for hand-written SQL**

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New Features

- **Annotations support**
- **Subselect fetching**
- **Property-level lazy fetching**
- **Support for update by ROWID on Oracle**
- **JMX-enabled statistics reporting**

XML Mappings

- **Map relational data directly to XML!**

This is an experimental feature of Hibernate 3.0.

- **Use persistent XML instead of Java objects**
- **Uses dom4j for manipulating XML**
- **Many applications including:**
 - data import/export**
 - externalization of entity data via JMS or SOAP**
 - XSLT-based reporting**

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Bulk update/delete by query

- **HSQL now supports 'delete', 'update' clauses**
- **Not much documentation (yet)**

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Virtualisation

- **”A unique, innovative feature for handling temporal, regional or permissioned data efficiently and elegantly in ORM.”**
- **Enable filters at the session-level**
- **Helps centralise query logic**
 - e.g. on logon, setup country filter – only see results for Australia**

Virtualisation

➤ Global Definition

➤ `<filter-def name="region">`
 `<filter-param name="regionCode" type="string"/>`
`</class>`

➤ Employee Mapping

➤ `<class name="Employee">`
 `...|`
 `<filter name="region" condition="REG = :regionCode"/>`
`</class>`

➤ Java Code

➤ `session.enableFilter("region").setParameter("regionCode", "AU");`
`List regionEmployees = session.createQuery("from Employee").list();`

Single-object multiple-table mapping

- **<join> mapping**
- **Map properties to other tables**
- ```
<class name="Person" table="PERSON">
 <id name="id" column="PERSON_ID">...</id>
 <join table="ADDRESS">|
 <key column="ADDRESS_ID"/>
 <property name="address"/>
 <property name="zip"/>
 <property name="country"/>
 </join>
```
- **Works well with components, lazy properties!**

# Support for hand-written SQL

- **Override Hibernate's SQL as needed**
- **Allows use of stored procedures**
- ```
<class name="Person">  
  <id name="id">  
    <generator class="increment"/>  
  </id>  
  <property name="name" not-null="true"/>  
  <sql-insert>INSERT INTO PERSON (NAME, ID) VALUES ( UPPER(?), ? )</sql-insert>  
  <sql-delete callable="true">{? = call deletePerson (?)}</sql-delete>  
  <sql-delete>DELETE FROM PERSON WHERE ID=?</sql-delete>  
</class>
```
- **NB: Parameters must be in order expected by Hibernate**

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Annotations support

- **Seperate tool – still in beta**
- **Annotations supplement config file
config file overrides!**
- **Supports EJB3 (JSR220) annotations**

Annotations – EJB Example

➤ *@Entity(access = AccessType.FIELD)*
public class Customer implements Serializable {
 @Id
 Long id;
 String firstName;
 String lastName;
 Date birthday;
 @Transient
 Integer age;
 @Dependent
 private Address homeAddress;
 @OneToMany(cascade=CascadeType.ALL, targetEntity="Order")
 @JoinColumn(name="CUSTOMER_ID")
 Set orders;
 // Getter/setter and business methods
}

Subselect fetching

- **Subselect – select within a select**
- **Equivalent to selecting from a view**
- **Good for collections, read-only objects**

```
➤ <class name="Summary">
  <subselect>
    select item.name, max(bid.amount), count(*)
    from item
    join bid on bid.item_id = item.id
    group by item.name
  </subselect>
  <synchronize table="item"/>
  <synchronize table="bid"/>
  <id name="name"/>
  ...
</class>
```

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Property-level lazy fetching

- **Properties can be marked lazy**

Similar to lazy collections

- **Only fetched when referenced**

Requires open session

- **Good for:**

components

multi-table objects

expensive properties (e.g. BLOBs)

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Support for update by ROWID on Oracle

- **More efficient updates for Oracle.**

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JMX-enabled statistics reporting

- **Hibernate property: `hibernate.generate_statistics`**
- **See Javadoc for `org.hibernate.stat` for full list**

http://www.hibernate.org/hib_docs/v3/api/org/hibernate/stat/Statistics.html
- **Can find out such stats as:**
 - number of cache hits/misses**
 - execution times of queries**
 - number and duration of sessions**
- **Can also browse the contents of the second-level cache.**

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Migration Guide

- **Hibernate 3.0 is *not* a drop-in replacement**
 - Not source-code compatible**
 - Hibernate 3.0 is *not* a drop-in replacement**
 - 3.0 and 2.1 systems can sit side-by-side**
- **New base package: `org.hibernate`**
- **`HibernateException` is now unchecked**
- **Convenience methods on `Session` removed:**
 - `find`, `iterate`, `filter`, `delete`**

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Migration Guide

- **Lifecycle, Validatable interfaces gone**
use Interceptor, event framework instead
- **UserTypes are more flexible**
Can accept parameters
- **PersistentEnum removed.**
- **Blob/Clob now wrapped**
Can unwrap if needed

Migration Guide

➤ **Collections are now lazy by default**

➤ **Some HSQL not valid anymore**

HSQL 1.1 'from f in class bar.Foo'

➤ **Uses ANTLR for HSQL parsing**

Not directly compatible with WebLogic!



Questions???